

YOURSTORY



snowflake

H2C



CoCo CLI

HACKATHON

GCC Edition

SupplyChainIQ

Governed Conversational Supply Chain Analytics on Snowflake CoCo & Cortex



"The LLM understands the question; the governed ontology decides what it means and where the answer comes from."

1. PROBLEM BRIEF

The Meaning Crisis in Supply Chain

- **Data, Not Meaning:** Teams have abundant data across ERP, WMS, and TMS, but inconsistent definitions yield 3 different answers to the same question.
- **Crisis Scenario:** Port of Houston bottleneck traps lithium-ion battery modules (PART_BAT_402), placing Gigafactory Texas at 3.8 days to stockout.
- **Personas Addressed:** Procurement VP (liquidated damages), Logistics Director (OTIF & demurrage), Plant Manager (assembly buffer runway).

2. ARCHITECTURE & COCO

Deterministic Bottom, Intelligent Top

- **Pipelines (CoCo):** 3 near real-time Dynamic Tables (1m lag) on COMPUTE_WH aggregating OTIF, DOI, and active disruptions.
- **Governed Semantics:** First-class METRIC_REGISTRY + semantic YAML model defining canonical metrics across personas.
- **Unstructured RAG:** Cortex Search Service (SUPPLIER_CONTRACTS_SEARCH) indexing vendor MSAs for SLA penalty clauses.
- **Action MCP Tools:** Registered tools for emergency air freight dispatch and formal Slack SLA breach alerts.

3. IMPACT STATEMENT

Measurable Outcomes & Scalability

- **100% Metric Consistency:** Guarantees mathematically identical numbers for OTD, DOI, and Landed Cost across all personas.
- **90% Faster Recovery:** Reduces contract breach discovery from 4 hours to seconds, recovering \$22,500 in liquidated damages.
- **Universal Scalability:** SCOR-compliant ontology readily adapts to Pharma, Retail, and Aerospace without changing pipelines.
- **Production Live:** Deployed natively as Streamlit in Snowflake (SiS) with full automated test validation suite.

Disparate ERP, logistics, and supplier systems yield conflicting answers, paralyzing cross-functional crisis response.

3 CONFLICTING ANSWERS

Metric Ambiguity & Silos

- **The 'Meaning' Breakdown:** Procurement, Planning, and Logistics calculate delivery performance using conflicting formulas and dates.
- **The Discrepancy:** When asked 'What is our on-time delivery?', Procurement gets 87%, Planning gets 82%, and Logistics gets 91%.
- **No Shared Ontology:** Missing relational model connecting Supplier → Part → Plant → Shipment → Order.

3.8 DAYS RUNWAY

EV Battery Assembly Crisis

- **Port Bottleneck Disruption:** Port of Houston maritime berth congestion traps critical battery cells (PART_BAT_402) from Apex Battery Cells Ltd.
- **Imminent Plant Stoppage:** Gigafactory Texas (Austin) burns 600 units/day with only 2,280 units on hand = exactly 3.8 Days of Inventory remaining!
- **Accumulating Demurrage:** Container demurrage charges at port already surge past \$19,500 USD.

\$22,500 PENALTY RISK

Cross-Domain Persona Needs

- **VP of Procurement:** Needs to calculate liquidated damages trapped in dense PDF Master Service Agreements (\$1,500/day after 3-day grace).
- **Logistics Director:** Needs real-time OTIF tracking, port demurrage monitoring, and carrier accountability.
- **Plant Manager:** Needs instantaneous stockout early warnings to trigger emergency air-freight re-routing.

A 4-tier governed data & AI pipeline built and orchestrated entirely via Snowflake CoCo CLI.

TIER 1: PIPELINES

Dynamic Tables (1m Lag)

- **7 Core Tables:** Seeded referentially sound data: DIM_SUPPLIERS, PARTS, PLANTS, ORDERS, SHIPMENTS, INVENTORY, CONTRACTS.
- **Near Real-Time DTs:** 3 Dynamic Tables continuously refreshing on 1-min lag on COMPUTE_WH.
- **Automated Rollups:** DT_SUPPLIER_PERFORMANCE, DT_PLANT_STOCKOUT_RISK, DT_ACTIVE_DISRUPTIONS.

TIER 2: SEMANTICS

Governed Metric Registry

- **Metric Registry:** First-class METRIC_REGISTRY cataloging canonical formulas, owners, and grains.
- **Semantic Model YAML:** supply_chain_semantic_model.yaml staged at @SEMANTIC_MODELS_STAGE.
- **Canonical Truth:** Centralized definitions for OTIF %, Days of Inventory, and Landed Cost.

TIER 3: CORTEX AI

Unstructured RAG Search

- **Cortex Search Service:** SUPPLIER_CONTRACTS_SEARCH indexes vendor MSAs using native vector embeddings.
- **Semantic Contract RAG:** Extracts Section 8.1 grace periods and Section 8.2 liquidated damages (\$1,500/day).
- **High Precision:** 0.68 cosine similarity and 1.24 reranker score on contract clauses.

TIER 4: AGENT & ACTION

CoCo & Action MCP Tools

- **Full CoCo Lifecycle:** Planning, SQL pipeline generation, execution, and automated testing suite.
- **MCP Action Tools:** expedite_purchase_order (air freight PO) & send_supplier_sla_breach_alert (Slack).
- **Streamlit in Snowflake:** Live interactive Command Center accessible directly in Snowsight.

Eliminating LLM hallucinations through intermediate query compilation, mathematical decomposition, and full auditability

ZERO HALLUCINATION Semantic Query Plan (JSON)

- **Intermediate Representation:** Natural language does NOT generate raw SQL directly. It compiles into a structured JSON query plan.
- **Entity & Path Grounding:** Specifies intent (performance_decomposition), metric (on_time_delivery), and traverses: Supplier → Part → Plant → Shipment.
- **Deterministic Boundary:** The LLM understands the question; the semantic view determines what the math means.

MATHEMATICAL ATTRIBUTION Deterministic 'Why' Engine

- **No Vague Explanations:** Explains why OTD dropped to 33.3% with exact mathematical component contributions.
- **Apex Battery Cells:** 4 delayed consignments contribute -66.7 pp, generating \$22,500 in accrued damages.
- **Port of Houston:** Shipments SHP_9002 & SHP_9011 stuck at berth, adding \$19,500 demurrage penalty fees.

AUDITABLE TRANSPARENCY The 3 Trust Buttons in SiS

- **[View Governed SQL]:** Provides immediate inspection of the verified SQL generated by the semantic layer.
- **[Metric Definition]:** Exposes metric owner (Logistics), grain (Shipment), and canonical SQL formula.
- **[Data Lineage]:** Traces full provenance from ERP → Dynamic Tables → Semantic View → Answer.
- **Live SiS App:** SUPPLY_CHAIN_COMMAND_CENTER deployed live in Snowsight.

Measurable Business Impact & Enterprise Extension

Proven ROI, universal SCOR-compliant industry adaptability, and production readiness beyond the hackathon.

QUANTIFIABLE ROI

Measurable Business Outcomes

- **100% Metric Consistency:** Eliminates cross-departmental conflicting reports across Procurement, Planning, and Logistics.
- **90% Faster Triage:** Cuts contract breach investigation from 4 hours of manual legal search to 3 seconds via Cortex Search.
- **\$22,500 Direct Recovery:** Automates liquidated damages penalty filings and averts a multi-million-dollar plant shutdown.

SCOR COMPLIANT

Universal Industry Scalability

- **Plug-and-Play Ontology:** The entity graph (Supplier → Part → Plant → Shipment → Order) is 100% industry-agnostic:
- **Pharmaceuticals:** API Suppliers → Vials → Formulation Plants → Cold-Chain Distribution → FDA SLA compliance.
- **Retail & E-commerce:** Vendors → SKUs → Regional Fulfillment Centers → Late Delivery Chargebacks.
- **Aerospace:** Fasteners → Aircraft Sub-assemblies → Hangars → FAA lead times.

BEYOND THE DEMO

Production Extensibility

- **Packaged CoCo Skill:** skills/supply_chain_agent_skill/SKILL.md ready for deployment across any Snowflake organization.
- **Unattended Automations:** Scheduled Snowflake Task (HOURLY_SUPPLY_CHAIN_MONITOR_TASK) runs 24/7 background anomaly checks.
- **GitHub Open Source:** All pipelines, tests, and documentation maintained at <https://github.com/chiraghs/CoCo-Learn>.